

Rahul Anand Nayanegali

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TECHNICAL SKILLS

AI Engineering: harness engineering (feedforward guides, feedback sensors); MCP, Claude Code, Kiro, OpenCode CLI; prompt engineering (CoT, Few-Shot)

Languages: TypeScript, JavaScript, Python, Java

Fullstack: Node.js, Flask, Spring Boot, REST APIs, GraphQL, gRPC, SSE, WebSockets, React, Next.js

Cloud & Infra: AWS (ECS, EC2, S3, Lambda, CloudFront), Docker, CI/CD

Testing: TDD, BDD, Contract Testing (Pact), Load Testing (k6), E2E (Playwright, Cypress), Unit & Integration (Jest, Supertest, Pytest)

EXPERIENCE

Software Development Engineer

Jun 2025 – Dec 2025

Amazon

Seattle, WA

- Co-owned Amazon Seller Central’s compliance platform, a business-critical system where UI regressions directly block seller listings and impact revenue; re-architected ActionWidget from a hardcoded, regulation-specific system to a schema-driven micro-frontend using the Strangler Fig pattern, onboarding new regulations incrementally with zero production downtime.
- Designed and built a **Java REST API** as a Backend for Frontend (**BFF**) layer for ActionWidget, consolidating calls to 3 internal Amazon APIs and translating aggregated responses into the widget’s data contract; offloaded orchestration logic from the frontend and reduced browser round trips from 3 to 1.
- Co-designed and built ActionWidget from scratch as a schema-driven, config-driven micro-frontend inspired by **hexagonal architecture**; decoupled regulation logic from rendering so new regulations onboard via config alone with no code change, replacing a hardcoded conditional-rendering system that blocked safe scaling.
- Established **TDD** practices across the team by authoring tests around API contracts, error handling, and edge cases; enforced through **trunk-based development** and code review gates, stabilizing production releases and reducing regression risk on revenue-critical seller flows.
- Designed AI-assisted **Playwright** test authoring pipeline using **BDD** specifications as LLM context, cutting test authoring time 63%, raising E2E coverage 60%, and accelerating delivery velocity 30%.
- Raised code review quality by enforcing single-purpose diffs, separating structural refactors from behavioral changes; smaller PRs surfaced potential bugs earlier, reducing regression risk during an active critical migration.

Software Engineer

Aug 2024 – Jun 2025

Shoptaki

New York, NY

- Engineered a Node.js service that automated text-to-SQL query execution for a natural language data interface; extracted live PostgreSQL schema context including table relationships, constraints, and join patterns, injected it alongside user queries into LLM prompts, and validated and sanitized generated SQL before execution against a staging database.
- Built the query interface in React and TypeScript to handle dynamically shaped result sets where columns and types were unknown at build time; implemented async execution states, error surfaces, and adaptive table rendering to handle unpredictable LLM output shapes.

Software Engineer

Feb 2021 – Jan 2023

CodeCraft Technologies

Bangalore, India

- Co-owned **React/TypeScript** component development for JioCinema on **Next.js 13** (Pages Router) with **Redux**; implemented **SSR** and code splitting via dynamic imports on streaming pages serving IPL and FIFA World Cup 2022 traffic.
- Modernized build pipeline for a high-traffic streaming platform (Webpack/Babel upgrade, asset optimization); cut build times 50% and unblocked ES6+ feature adoption.
- Migrated application state from **Redux** to **Apollo Client (GraphQL)**, implementing a unified GraphQL infrastructure layer; eliminated boilerplate and reduced bundle size 20%.
- Built and maintained backend services in **Node.js** and Python on **AWS** (ECS, EC2, Lambda, S3); engineered robust data processing pipelines.

Assistant Systems Engineer

Nov 2018 – Apr 2020

Tata Consultancy Services

Bangalore, India

- Built real-time IoT monitoring dashboard using **Angular**, Flask, and **Server-Sent Events (SSE)**, handling continuous sensor data streams at scale; designed RESTful API endpoints for auth, data retrieval, and live updates.
- Optimized frontend performance via lazy loading, image compression, and JS minification; delivered measurable load time improvements on production dashboard.

EDUCATION

Pace University

New York, NY

Master of Science in Computer Science

Jan 2023 – May 2024

Sri Siddhartha Institute of Technology

Tumkur, India

Bachelor of Engineering in Electronics and Communication Engineering

September 2018